

9.2 The impact of globalisation and the role of transnational corporations

This chapter will explain:

- ★ globalisation and its key features
- ★ the impacts of globalisation on trade, transport, culture, communications and technology
- ★ the role, location and costs and benefits of transnational corporations
- ★ detailed specific example: globalisation and TNCs in Mongolia.

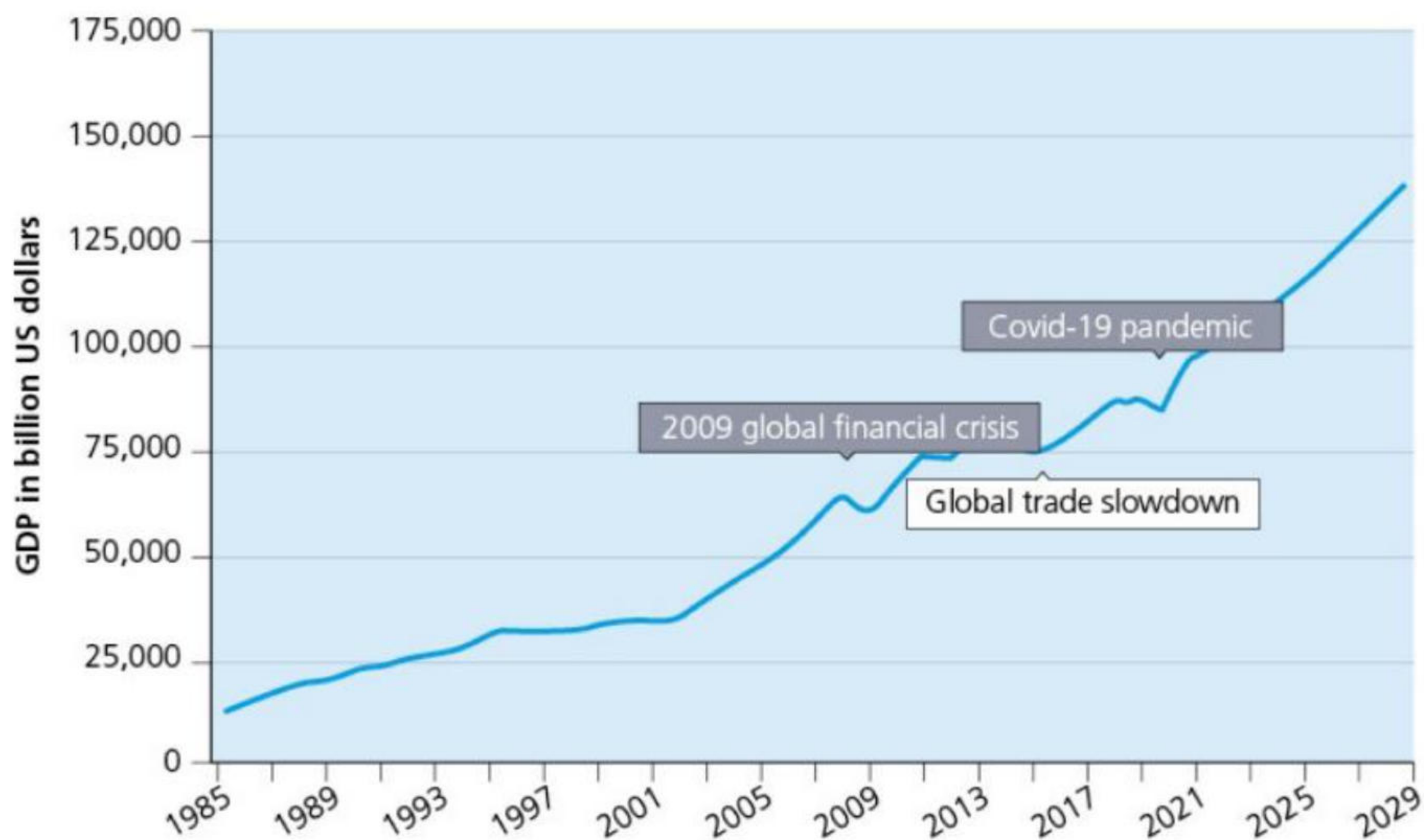
Globalisation and its key features

'**Globalisation**' is a term that describes the rapidly increasing and more complex global links that have developed since the 1960s. These links are economic, cultural and political. The global economy today is more extensive and complicated than it has ever been. The key features of globalisation are summarised in [Figure 9.22 \(page 221\)](#). A hierarchy of **global cities** has emerged to act as the command centres of the global economy. New York, London and Tokyo are at the highest level in this hierarchy. These are major financial and decision-making centres. The number of global cities has increased as the process of globalisation has advanced.

Competition within and between the different levels of the global hierarchy is intensifying because of the enormous wealth that such business and financial centres can create. Expanding global connectivity has encouraged the growth in production of goods and services in many countries. Attracting more business creates jobs and wealth. As a result, the global economy has expanded significantly in recent decades. It was valued at \$100 trillion in 2022, an all-time record ([Figure 9.19](#)).

[Figure 9.19](#) shows that at times severe global economic shocks have interrupted the growth of the global economy. In recent decades these have been:

- » the 2008–09 global financial crisis
- » the Covid-19 pandemic.

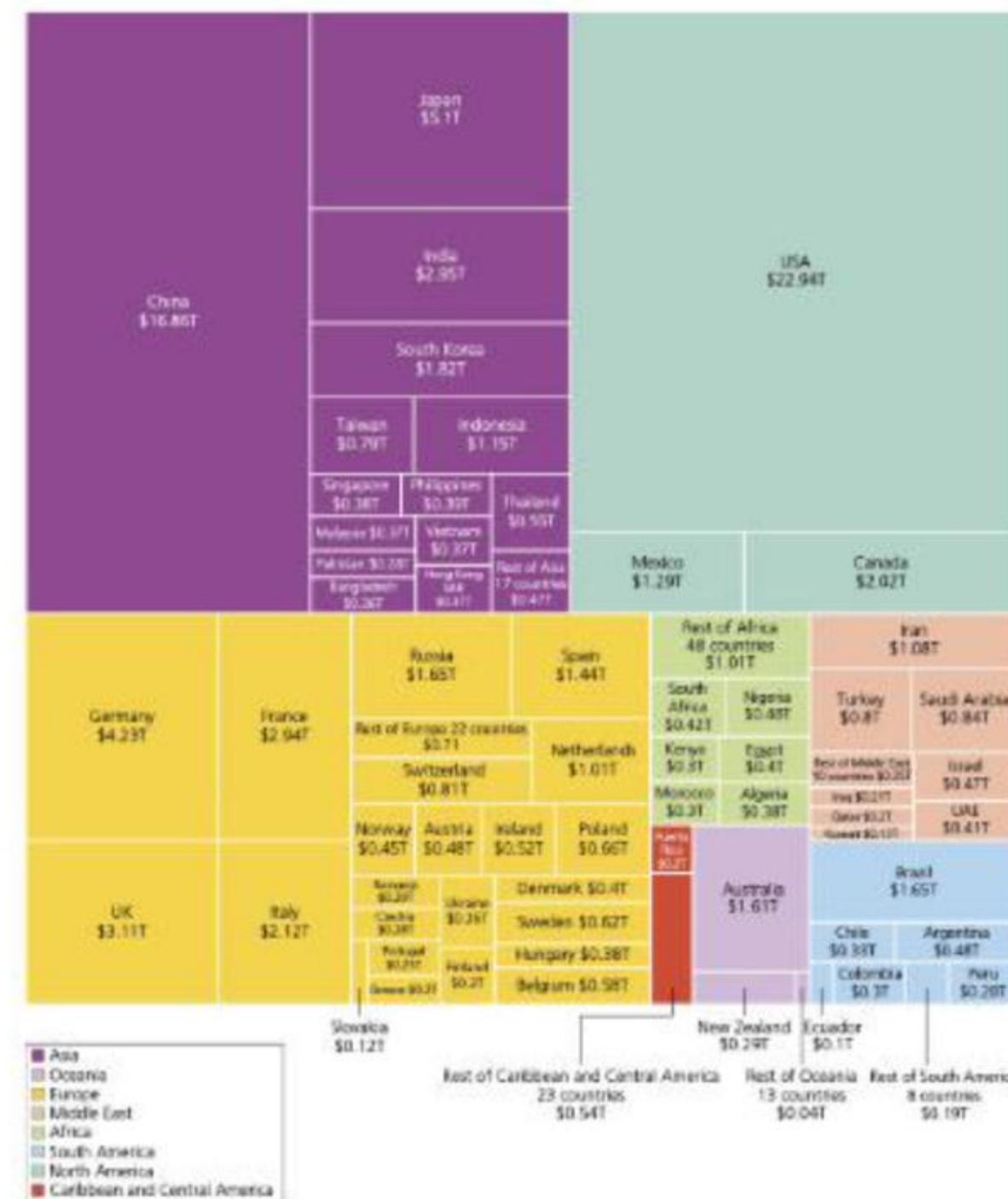


▲ **Figure 9.19** Global GDP at current price, 1985–2022, and estimates to 2029

Spatial distribution

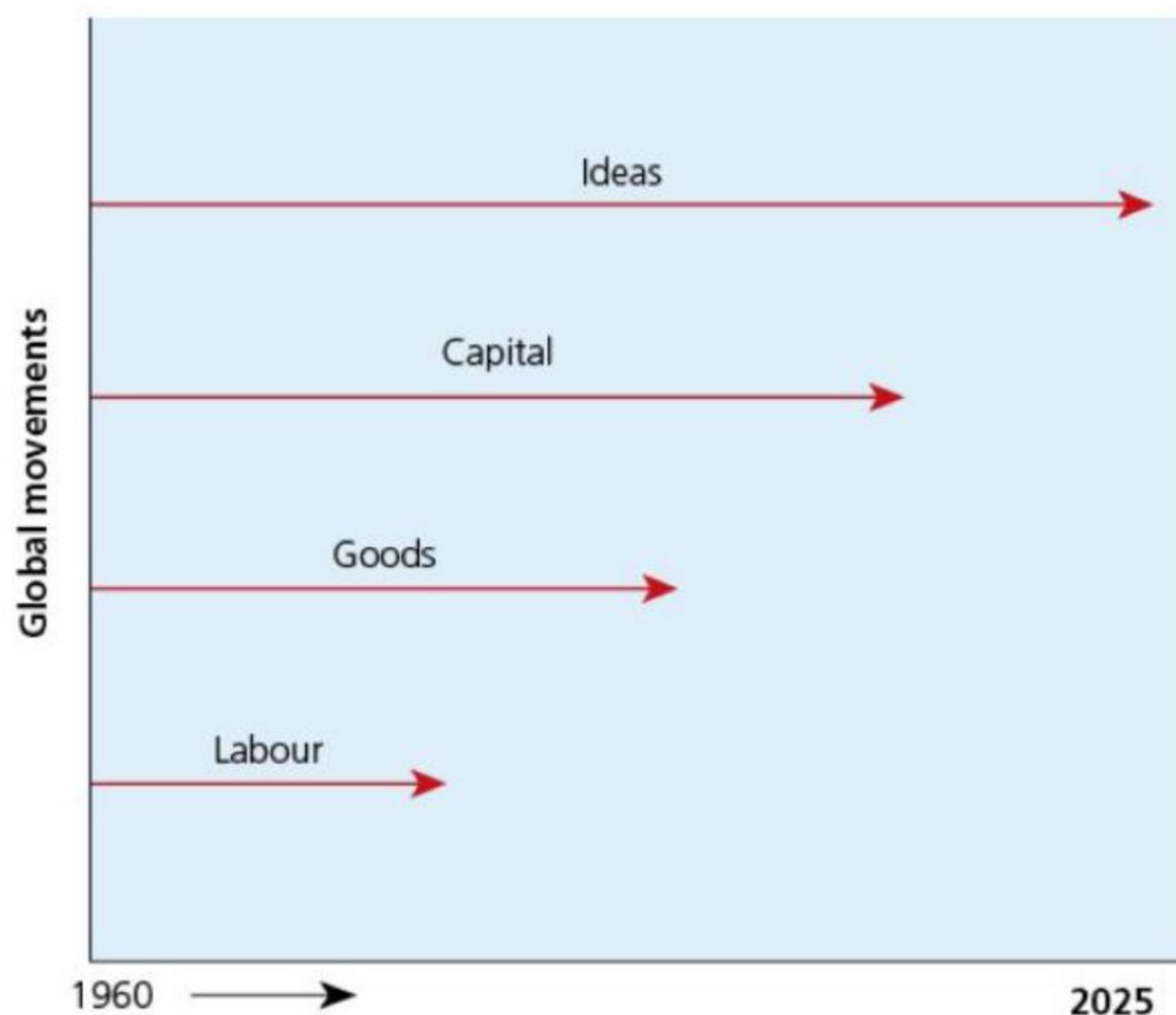
[Figure 9.20](#) illustrates the spatial distribution of the global economy in 2021. In this year the USA and China together accounted for more than 42 per cent of global GDP. The USA has been the world's largest economy since 1871. However, by 2030, China's GDP is projected to surpass that of the USA. The economic rise of China and other large emerging economies has been rapid. Asia is now the largest continental economy in the world, accounting for 39 per cent of world GDP in 2021, up from 15.1 per cent in 1970. The rapid growth of emerging economies has brought about a changing world economic order. Some countries have benefited much more than others in this process. Many of the world's smallest economies are located

in the Oceania region. These are small Pacific island states with very limited resources.



▲ **Figure 9.20** The spatial distribution of the global economy, 2021

Most political borders are not the obstacles they once were, so goods, capital, labour and ideas flow more freely across them than ever before ([Figure 9.21](#)). While the movements of all of these elements have increased significantly, the barriers they face differ in strength. The ‘movement’ facing the most formidable barriers is labour, because of the restrictions that governments put in place to limit international migration.



▲ **Figure 9.21** The ability to overcome political borders in the globalisation era

Factors encouraging the globalisation of economic activity

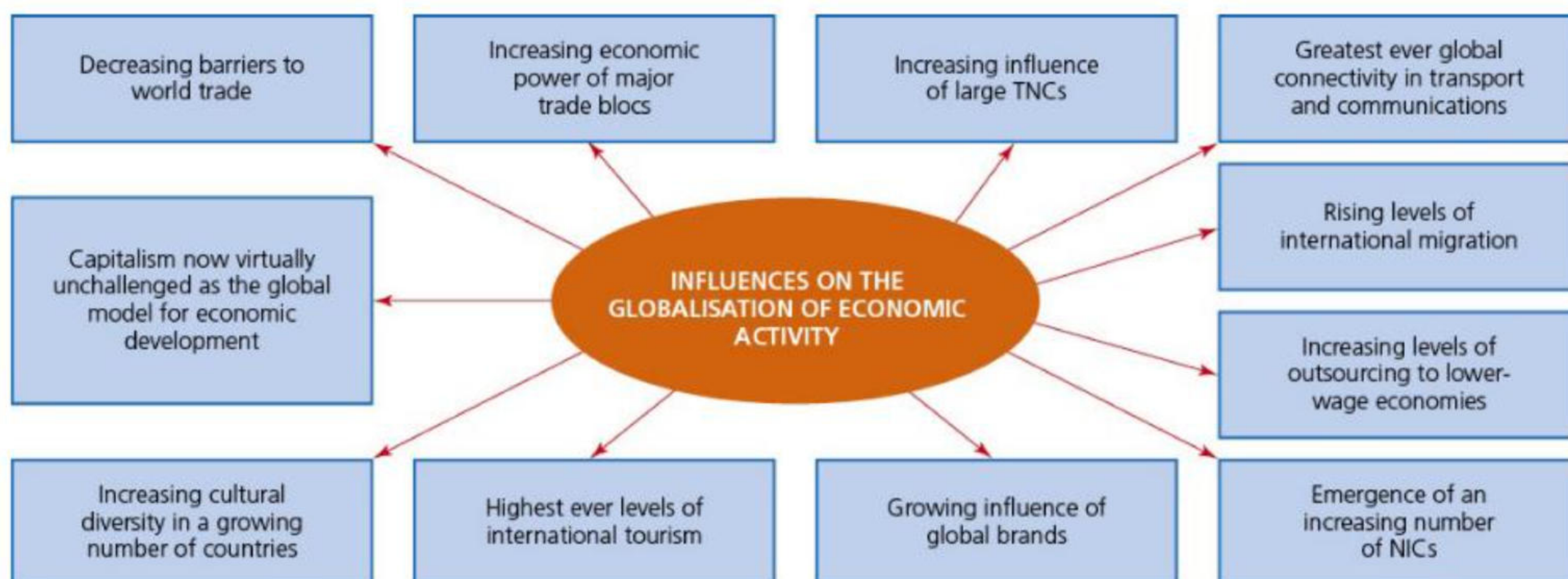
[Figure 9.22 \(page 221\)](#) shows the main influences on the globalisation of economic activity. Until the post-1950 period, industrial production was mainly organised within individual countries. This has changed rapidly in the last 70 years or so, with the emergence of a **new international division of labour (NIDL)**, which has developed hand in hand with the global increase in foreign direct investment. The NIDL divides production into different skills and tasks that are often spread across a number of countries. The following are other significant factors responsible for economic globalisation:

- » The increasing complexity of international trade flows as the NIDL has developed.
- » Major advances in trade liberalisation under the World Trade Organization. The barriers to world trade (tariffs, quotas and regulations) are much lower today than in the past. This means there is more incentive to trade.
- » The emergence of free-market governments in the USA and UK around 1970. The economic policies, such as privatisation, that were developed by these governments influenced policy-making in many other

countries.

- » The emergence of an increasing number of newly industrialised countries (emerging economies).
- » The integration of the former Soviet Union and its Eastern European satellites into the capitalist system after the fall of communism in the late 1980s. Now no significant group of countries stands outside the free-market global system.
- » The opening up of other economies, particularly those of China and India, as these countries wanted to benefit from the process of globalisation.
- » The deregulation of world financial markets, allowing a much greater level of international competition in financial services.

However, there are concerns that the era of globalisation has at the very least come to a period of stagnation, and some writers assert that a level of deglobalisation is already occurring. This is because of the growing geopolitical tensions between the Western economies and China and Russia.



▲ **Figure 9.22** Factors encouraging the globalisation of economic activity

Labour and labour migration

The large pools of relatively low-cost labour in emerging and developing countries have been a major attraction for transnational companies to manufacture in such countries. The increasing skills of these pools of labour often set off a cycle of investment, attracting production higher up the value chain. As the cost of labour has risen in the fastest-growing emerging and developing countries, companies have often looked to relocate to lower-wage economies, for example from China to Vietnam in recent years.

Fast-growing economies often experience shortages of workers. This situation can be at least partially solved by labour migration. Potential employees in lower-income countries are attracted to the jobs available in higher-income countries. The international mobility of highly skilled workers has increased substantially since the 1990s. TNCs often stress the importance of being able to attract highly skilled workers from around the world.

Activities

- 1 Define globalisation.
 - 2 What is a global city?
 - 3 Explain the new international division of labour (NIDL).
 - 4 What is foreign direct investment?
 - 5 Discuss three factors (other than those covered in the previous activities) encouraging the globalisation of economic activity.
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The impacts of globalisation

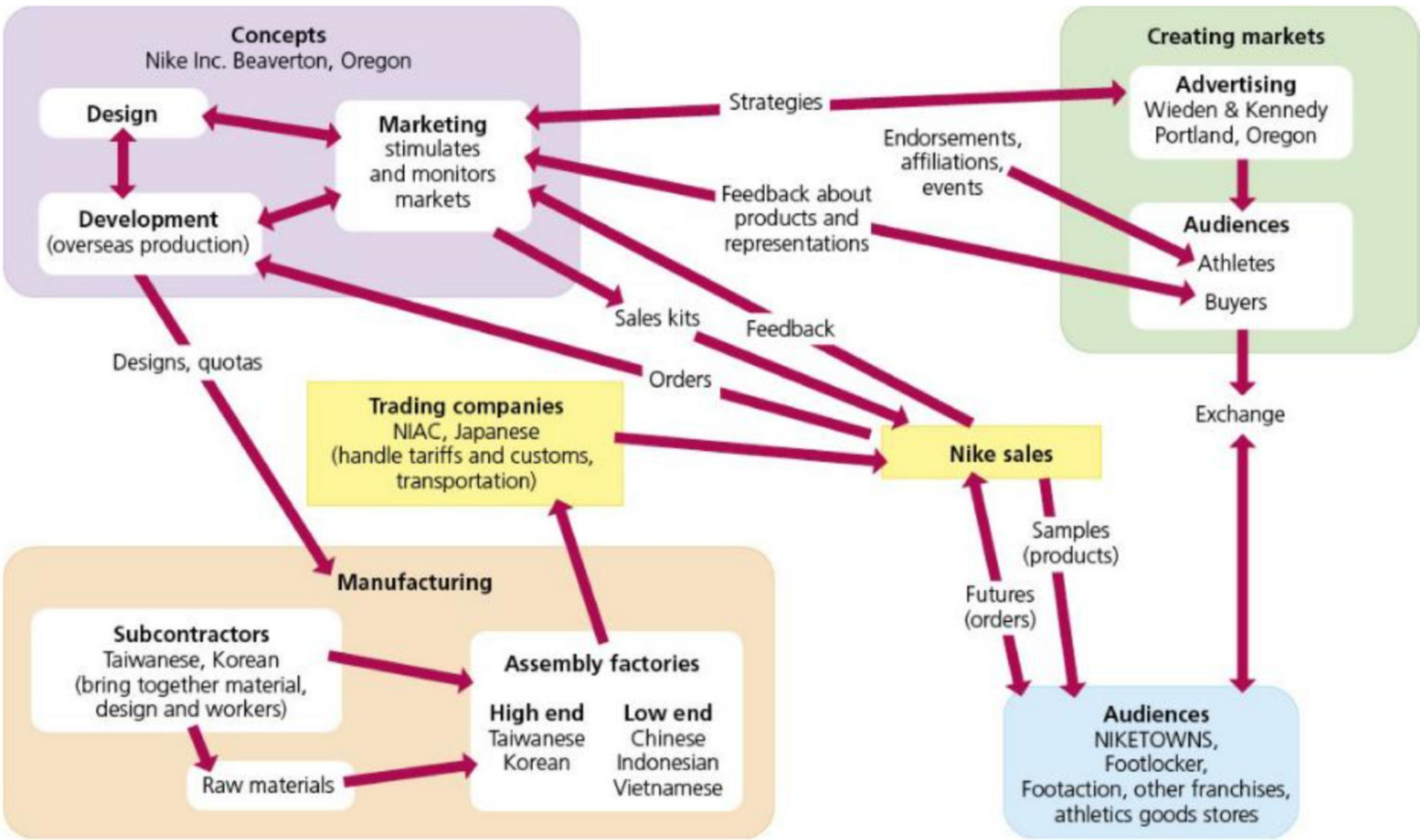
Trade

The development of **commodity chains** (production chains) has been an important aspect of globalisation. Each stage of manufacturing is usually completed in the location where production costs are lowest. Value is added at each stage of the process. An example is the production of footwear by companies such as Nike and Adidas. [Figure 9.23](#) illustrates Nike's international linkages.

The barriers to **international trade** (tariffs, quotas, regulations) are much lower today than in the past. The World Trade Organization (WTO) and its predecessor the General Agreement on Tariffs and Trade (GATT) have chaired many rounds of international trade talks to reduce the barriers to trade. Between 1945 and 2021, there was a 40-fold increase in international trade. In 2023, global trade was valued at \$31 trillion, down from an all-time high of \$32 trillion in 2022.

The development of **trade blocs** has been an important part of this process. A trade bloc is a group of countries that share trade agreements between each other. Since the 1950s, there have been many

examples of groups of countries joining together to stimulate trade between themselves and to obtain other benefits from economic cooperation. The following forms of increasing economic integration between countries can be recognised: free trade areas, customs unions, common markets and economic unions.



▲ **Figure 9.23** Nike’s commodity linkages

Longstanding examples of trade blocs include the European Union (EU) and the North American Free Trade Agreement (NAFTA). The most recent example is the African Continental Free Trade Area, which was formed in 2018 and is comprised of 54 nations. Apart from trade blocs, there are a number of looser trade groupings aiming to foster the mutual interests of member countries.

There is a strong relationship between trade and development. In general, countries that have a high level of trade are richer than those with lower levels of trade. Countries that can produce goods and services that are in demand elsewhere in the world will benefit from strong inflows of foreign currency and from the employment their industries provide. Foreign currency allows a country to purchase goods and services from abroad that it either does not produce itself or does not produce in large enough quantities.

An important economic grouping is the Group of Seven (G7), which holds meetings on a regular basis to discuss all aspects of the global economy. The members of G7 are the USA, Canada, Japan, the UK, Germany, France and Italy. A very significant and more recent change was the establishment of the Group of 20 (G20) in the wake of the 1997 Asian financial crisis, to bring together advanced and emerging economies to stabilise the global financial market. Every continent is represented in the G20 group. However, smaller countries still feel they have little say in policy formation. African countries have noted that the continent has only one country (South Africa) representing it.

Advances in transportation

Major advances in transportation have reduced the geographical barriers separating countries and peoples. Transport systems are the means by which materials, products and people are transferred from place to place. **Transport hubs** are crucial for global connectivity and economic growth. The expansion of international trade has created global supply chains that rely on efficient transportation and logistics systems. The world’s largest transport hubs are massive global operations ([Table 9.4](#)).

▼ **Table 9.4** The largest transport hubs in the world: air, sea and ground

Location	Characteristics
Air: Hong Kong International Airport	● High-tech facilities and technology, such as automated storage and retrieval systems, temperature-controlled storage and advanced cargo-tracking systems ● The location of over 80 cargo carriers, creating a network to over 220 destinations worldwide. In 2023, the airport processed more than 4.3 million tonnes of cargo ● It is a key centre for shipments between Asia and the rest of the world
Sea: Port of Shanghai, China	● In 2021, the port processed over 49 million TEU (twenty-foot equivalent units) of cargo in containers, which was approximately one-third of China’s total container throughput

	● It has links to more than 2000 ports in over 160 countries ● It benefits from the most advanced technology for handling, tracking and distribution
Ground: FedEx World Hub, Memphis, Tennessee, USA	● FedEx is the world's largest cargo airline in terms of both fleet size and freight tonnes flown ● Located at Memphis International Airport (with an area of 3.5 km²), it is the main centre for the FedEx Express delivery network, which connects to over 220 countries and territories globally ● It processes millions of packages and documents daily, with sophisticated equipment, such as automated guided vehicles, advanced scanning and tracking systems, and real-time data analytics. This ensures the smooth running of operations throughout the site

There have been significant developments in logistics, with the aim of making the shipment of freight from one form of transport to another as seamless as possible. Digital and smart transportation systems have advanced efficiency and safety, and have been instrumental in the rapid development of e-commerce and online marketplaces. E-commerce has resulted in the development of new delivery models, such as same-day delivery.

Air freight has increased in importance in recent decades, particularly for high-value and time-sensitive goods such as electronics, pharmaceuticals and perishable goods. However, most goods are still carried by sea. In many countries, deep-sea ports have been built to accommodate ultra-large container ships, which allow for the most extensive economies of scale.

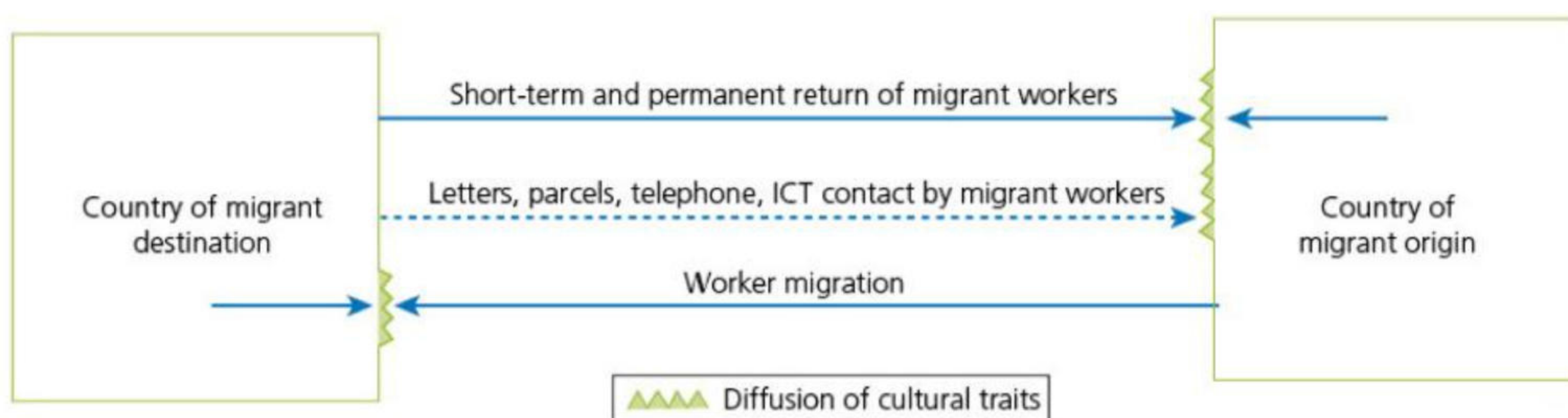
Autonomous vehicles may well become significant in moving freight as well as people in the not too distant future. An autonomous vehicle is one that can drive itself from a starting point to a predetermined destination in an 'autopilot' mode using various in-vehicle technologies and sensors. Significant developments in this area could transform transportation and further integrate global markets.

However, while the pace of transport infrastructure development has been stunning in many HICs and emerging economies, many LICs still face the challenge of severe infrastructure limitations. Upgrading infrastructure to a high standard is an expensive process, and the global gap in infrastructure capability is widening rather than narrowing for many LICs. This will further hinder their connectivity and integration into the global economy.

Culture

The mixing of cultures is a major dimension of globalisation. **Cultural diffusion** is the process of the spreading of cultural traits from one place to another. Cultures can move in both real space and cyberspace. This has occurred through:

- » migration, which circulates ideas, values and beliefs around the world and plays a significant role in cultural diffusion ([Figure 9.24](#))
- » the rapid spread of new ideas and fashions through the mass media, trade and travel
- » the growth of global brands, such as Coca-Cola and McDonald's, which serve as common reference points
- » the internet, which has allowed individual and mass communication on a scale never available before
- » the transport revolution, which has facilitated the mass movement of people and products around the world.



▲ **Figure 9.24** Worker migration and cultural diffusion

The spread of a global **consumer culture** has been important to the success of many major companies. The mass media have been used very effectively to encourage consumers to 'want' more than they 'need'. The power of brands and their global marketing strategies cannot be underestimated. This is particularly so for food, beverages and fashion.

Globalisation has undoubtedly had a significant impact on the increasing uniformity of landscapes. [Figure 9.25](#) illustrates the main ways in which common urban characteristics have diffused around the world. Many developments will of course encompass local traits as well, but the strong global elements will be clear to see. Industrial, agricultural, touristic and transport landscapes have also become more uniform as similar processes worldwide have influenced their development. Some people find this disappointing as more and more places lose their 'uniqueness'.



▲ **Figure 9.25** Aspects of global urban uniformity

Communications

Communications systems are the ways in which information is transmitted from place to place in the form of ideas, instructions and images. As time has progressed, the diffusion of new ideas has speeded up so that a technical breakthrough in one part of the world can have an impact on other parts of the world much more quickly than ever before.

Major advances in telecommunications systems have significantly improved global connectivity and have influenced all aspects of economic activity. The internet has been essential to the development and speed of globalisation. It is the fastest-growing mode of communication ever. It has been argued that the massive advances in information and communications technology over the last 40 years or so have been the main driver of globalisation.

The number of internet users around the world increased from 361 million in 2000 to over 5 billion in 2024. According to [Statistica.com](https://www.statista.com), as of April 2024 there were 5.44 billion internet users worldwide – 67.1 per cent of the global population. Of this total, 5.07 billion people were social media users.

The internet has allowed companies to manage complex operations all over the world, and to talk directly to their customers. Companies can react more quickly than ever before to changing consumer demand.

Technology

Advances in technology have affected all aspects of global economic activity. Robotics technology is gathering pace. Examples include driverless cars and drones. The impact on business organisations and many sectors of employment will be considerable.

According to the International Federation of Robotics (IFR), the top five global robotics trends in 2024 are as follows:

- » Artificial Intelligence (AI) and machine learning: Robot manufacturers are developing technology that allows users to program robots using natural language instead of code.
- » Robots expanding to new applications: Advances in sensors, vision technologies and smart grippers allow robots to respond in real time to changes in their environment to work safely alongside human workers.
- » Mobile manipulators: Automating material-handling tasks in industries such as automotive, aerospace and logistics.
- » Digital twins: The twin exists purely as a computer model, which can save costs by being stress-tested and modified with no safety implications.
- » Humanoid robots: The human-like design allows these robots to work flexibly in environments that were created for humans.

China aims to mass-produce humanoid robots by 2025. The Chinese Ministry of Industry and Information Technology (MIIT) predicts humanoids will become a **disruptive technology**, like computers and smartphones did before them. This could transform the production of goods, as well as the way that human societies live.

Activities

- 1 Write a brief summary of the linkages illustrated by [Figure 9.23 \(page 222\)](#).

- 2 List the barriers to international trade.
- 3 Explain the processes illustrated by [Figure 9.24 \(page 224\)](#).
- 4 Find out more about one of the robotic trends listed above. Give a brief class presentation of your research.

Transnational corporations

Transnational corporations (TNCs) and nation states (countries) are the two main elements of the global economy. The governments of countries individually and collectively (through global institutions) set the rules for the global economy, but the bulk of investment is through TNCs. Thus, TNCs have a substantial influence on the global economy in general, and on the countries in which they choose to locate in particular. [Table 9.5](#) shows the ten largest global corporations by revenue. The list is dominated by the USA and China, which each have four of the top ten global corporations.

▼ **Table 9.5** The ten largest global corporations by revenue

Rank	Company	Country	Revenues (US\$ million)	Profits (US\$ million)
1	Walmart	USA	572,754	13,673
2	Amazon	USA	469,822	33,364
3	State Grid	China	460,617	7138
4	China National Petroleum	China	411,693	9638
5	Sinopec Group	China	401,314	8316
6	Saudi Aramco	Saudi Arabia	400,399	105,369
7	Apple	USA	365,817	94,680
8	Volkswagen	Germany	295,820	18,187
9	China State Construction Engineering	China	293,712	4444
10	CVC Health	USA	292,111	7910

Interesting note

Walmart, the US retail giant, has been the world's largest company by revenue since 2014. It operates a chain of hypermarkets, discount department stores and grocery stores in the USA and 23 other countries, employing 2.1 million people around the world.

The role of TNCs

TNCs play a major role in world trade in terms of what and where they buy and sell. A considerable proportion of world trade is intra-firm, taking place within TNCs. The organisation of the major car manufacturing companies ([Figure 9.26](#)) exemplifies **intra-firm trade**, with engines, gearboxes and other key components produced in a number of countries and exported for assembly elsewhere. Apart from their direct ownership of productive activities, many TNCs are involved in a web of collaborative relationships with other companies across the globe.



▲ **Figure 9.26** Japanese investment in the UK: Suzuki in Crawley

The increase in the importance of **global value chains (GVCs)** in recent decades has become a major feature of economic globalisation. This is the trend to sub-divide the value chain to produce goods in stages in a number of locations, adding value at each stage. The overall objective is to produce the final product at the lowest total cost.

TNCs are involved in all economic sectors (primary, secondary, tertiary and quaternary). The 100 largest TNCs represent a significant proportion of total global production. Manufacturing industry at first, and more recently services, have relocated in significant numbers from developed countries to selected emerging and developing countries as TNCs have taken advantage of lower labour costs and other ways to reduce costs. This process has resulted in the emergence of an increasing number of newly industrialised countries (emerging countries) since the 1960s. Major examples are China, India and Brazil. The share of emerging economies in global foreign direct investment (FDI) inflows has risen significantly in recent decades.

TNCs are the main source of FDI. TNCs invest to make profits and are the driving force behind economic globalisation. Low- and middle-income countries often rely heavily on TNCs (which can be from both developed countries and emerging economies) to provide investment to develop large businesses and infrastructure. Countries generally welcome investment from TNCs, and there is often competition between countries for such investment. This is probably because many developing countries may not be able to attract other sources of investment, and feel that the advantages of hosting TNCs outweigh the disadvantages.

Such FDI has increased rapidly over the last 60 years. Many major global companies have operations in a large number of countries. For example, India's Tata Group has operations in more than 100 countries and receives more than 60 per cent of its revenue from outside India.

TNCs from emerging economies

Thirty years ago the vast majority of the world's TNCs had their headquarters in Northern America, Western Europe and Japan. However, since then, emerging economies have been accounting for an increasing slice of the global economy. Much of this economic growth has been achieved through the expansion of their own most important companies, first domestically (as national corporations) and more recently on an international basis (as TNCs).

The global organisation of TNCs

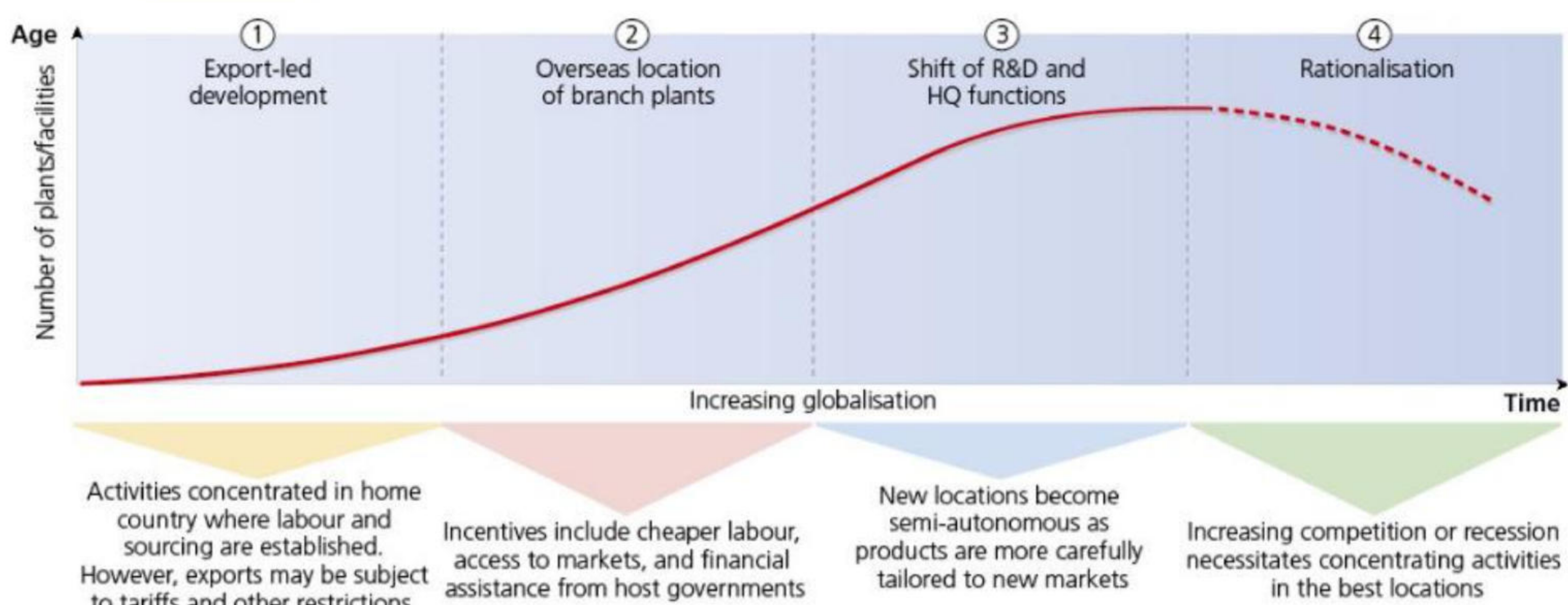
TNCs vary widely in overall size and international scope. Variations include:

- » the number of countries
- » the number of subsidiaries
- » the share of production accounted for by foreign activities
- » the division of research activities and routine tasks by country
- » the balance of advantages and disadvantages to the countries in which they operate.

Large TNCs often exhibit three organisational levels:

- » **Headquarters:** The headquarters of a TNC will typically be in the city where the company was established.
- » **Research and development:** This is likely to be located in the city where the company was established, or in other areas within the same country.
- » **Branch plants:** These are often the first to be located in other countries.

However, some of the largest and most successful TNCs have divided their business operations into world regions, each with its own research and development facilities and a high level of independent decision-making. [Figure 9.27](#) shows the locational changes that tend to occur as TNCs develop.



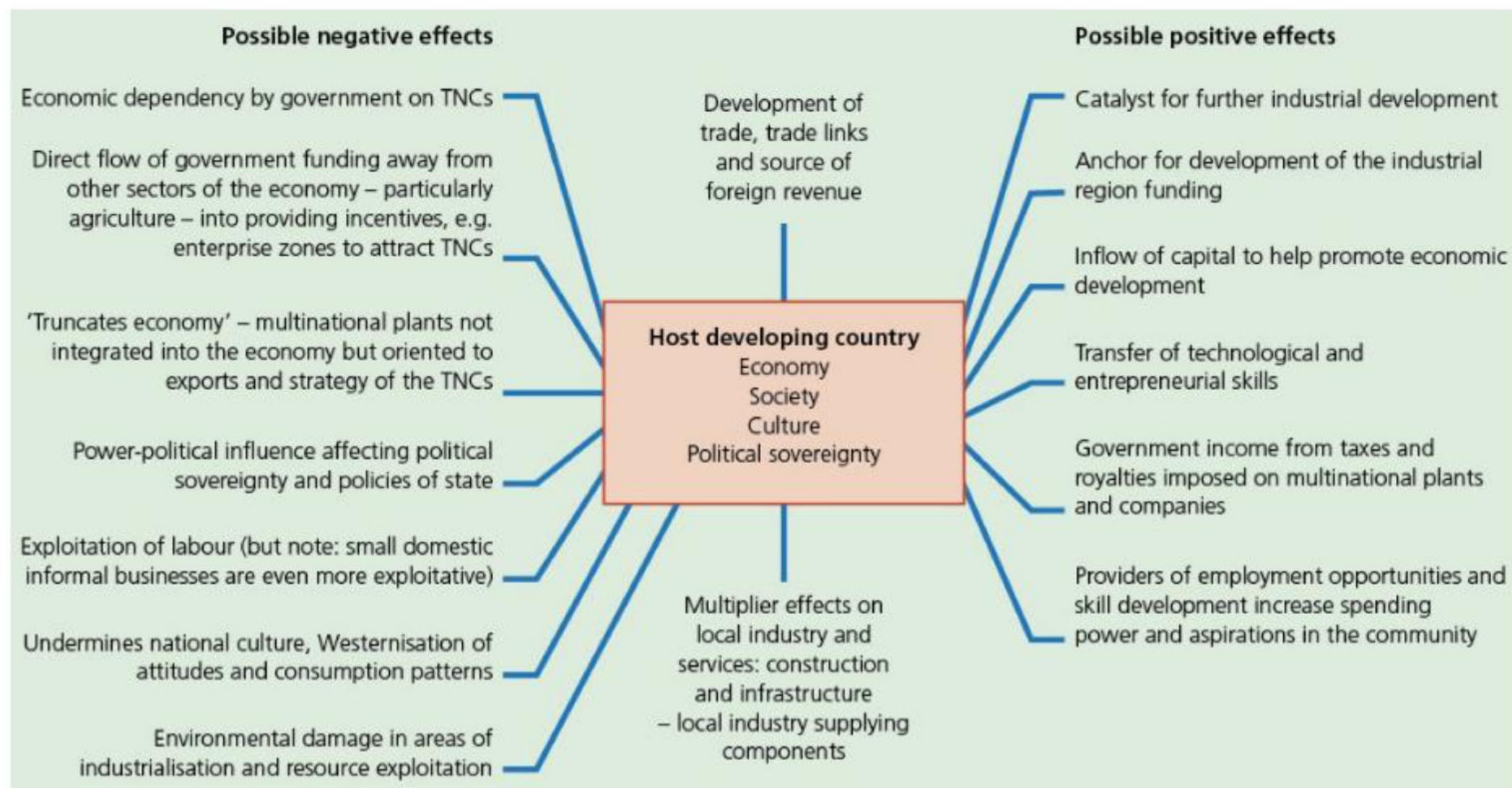
▲ **Figure 9.27** The development of TNCs: locational changes

The benefits and costs to countries hosting TNCs

There are many advantages for large companies in working at the global scale:

- » Sourcing raw materials and components on a global basis reduces costs.
- » TNCs can seek out the lowest-cost locations for labour.
- » Selling goods and services to a global market allows TNCs to achieve large economies of scale.
- » Global marketing helps to establish brands with huge appeal all around the world.

[Figure 9.28](#) summarises the possible positive and negative effects of TNCs locating in developing countries. The balance of issues will vary considerably in each individual case, and much depends on the ethical standards of individual companies. TNCs can of course also link developed nations (for example German companies operating in the USA) and emerging and developing countries (Chinese companies in South Africa).



▲ **Figure 9.28** The impact of transnational corporations on host developing countries

▼ **Table 9.6** The potential advantages and disadvantages of TNCs

Country	Possible advantages	Possible disadvantages
Headquarters in developed country	● Positive employment impact and stimulus to the development of high-level skills ● Direct and indirect contribution to local and national tax base	● Indirect loss of jobs and negative impact as products are imported ● Trade unions complain of an uneven playing field because of the big contrast in working conditions between developing and developed countries
Impact of outsourcing in developing country	● Creates substantial employment ● Pays higher wages than local companies ● Improves the skills base of the local population ● The success of a global brand may attract other TNCs, setting off the process of cumulative causation ● Exports are a positive contribution ● Sets new standards for home-grown companies ● Contribution to local tax base helps pay for improvements to infrastructure	● Concerns over the exploitation of cheap labour and poor working conditions ● Company image and advertising may help to undermine national culture ● Concerns about the political influence of large TNCs ● The knowledge that investment could be transferred quickly to lower-cost locations

TNCs exert power in host countries because of their size in terms of the number of people they directly and indirectly employ, the resources they use, and the capital flows they direct. The way TNCs operate has come under much criticism for some time now. This relates not just to the impact on developing economies, but also to the impact on the countries of origin of TNCs. [Table 9.6](#) summarises the case of the US sportswear firm Nike.

The anti-globalisation movement has questioned the activities of many TNCs and has stressed:

- » the way in which some TNCs have played off one country against another, seeking lower wages for workers and lower taxes from governments
- » the involvement of large TNCs in most of the world's serious environmental issues
- » the undue influence that some large corporations have on large international organisations.

Activities

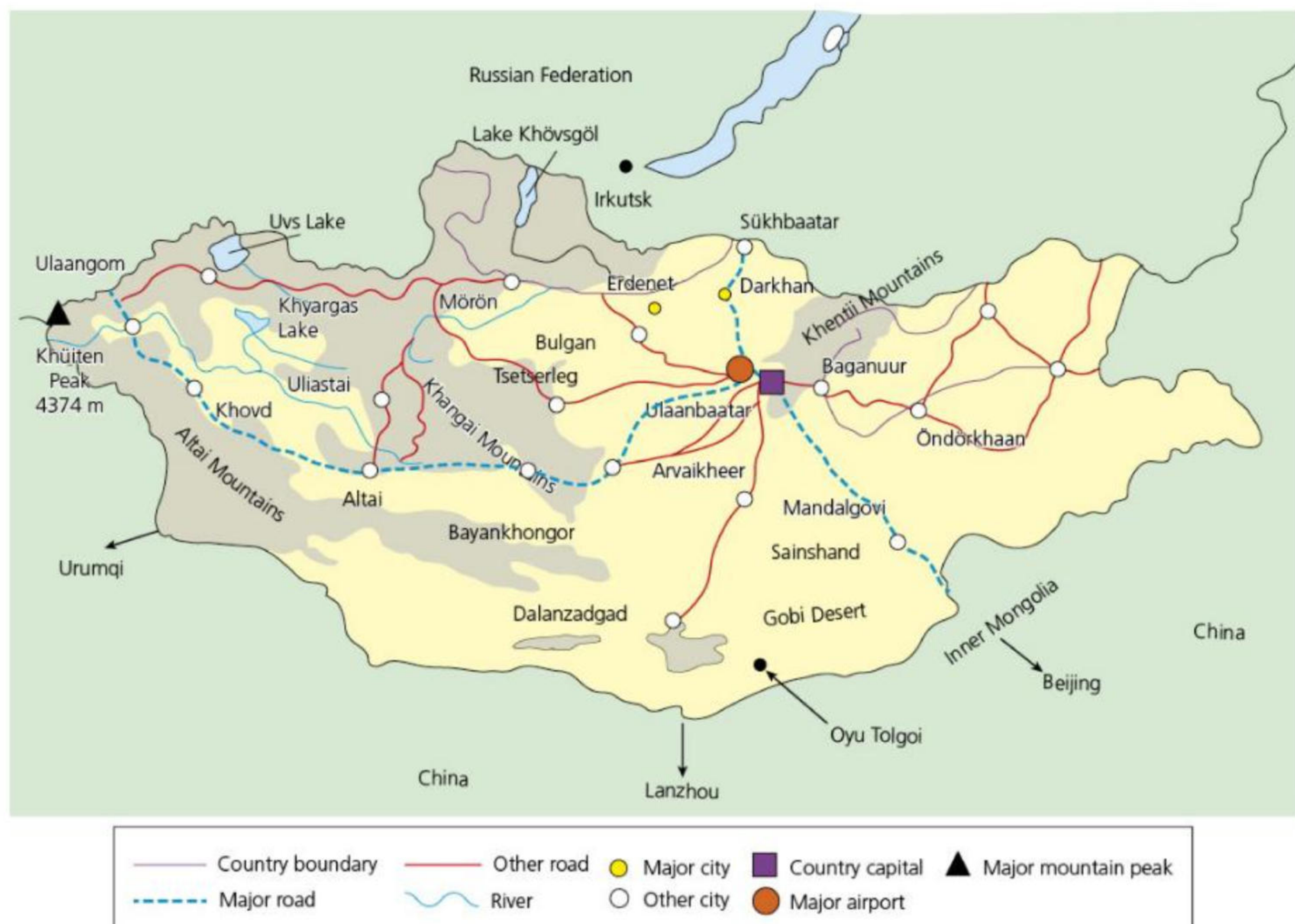
- 1 Define a transnational corporation.
- 2 Why have TNCs been so important in the process of globalisation?
- 3 What are global value chains and how are they related to TNCs?
- 4 Study [Figure 9.27 \(page 227\)](#). What are the locational changes often characteristic of the development of TNCs?
- 5 Look at [Figure 9.28 \(page 227\)](#). For developing countries, discuss two possible negative effects and two possible positive effects of hosting a TNC.

Detailed specific example

Globalisation and TNCs in Mongolia

Location and population

Mongolia is a landlocked country between Russia and China ([Figure 9.29](#)). It covers 1.5 million km², an area six times the size of the UK. The country has a 4670 km border with China and a 3485 km border with Russia. About 80 per cent of the country lies above 1000 metres. However, it has a population of only 3.3 million. Mongolia has experienced rapid urbanisation. Over 1.5 million people now live in Ulaanbaatar (UB), which is the 'coldest' capital city in the world. However, it is the only urban area in Mongolia with over 100,000 residents. Few countries exhibit such a high level of urban primacy. This is when a country's main city experiences excessive population concentration and development, often at the expense of other parts of the country.



▲ **Figure 9.29** A map of Mongolia

The rest of the population is spread over vast, often inaccessible areas. Here, movement associated with herding is the norm. Mongolia has the lowest national population density in the world, at 2.13 persons per square kilometre. Funding infrastructure in such a large and sparsely populated country is a significant development problem.

The impacts of globalisation on Mongolia

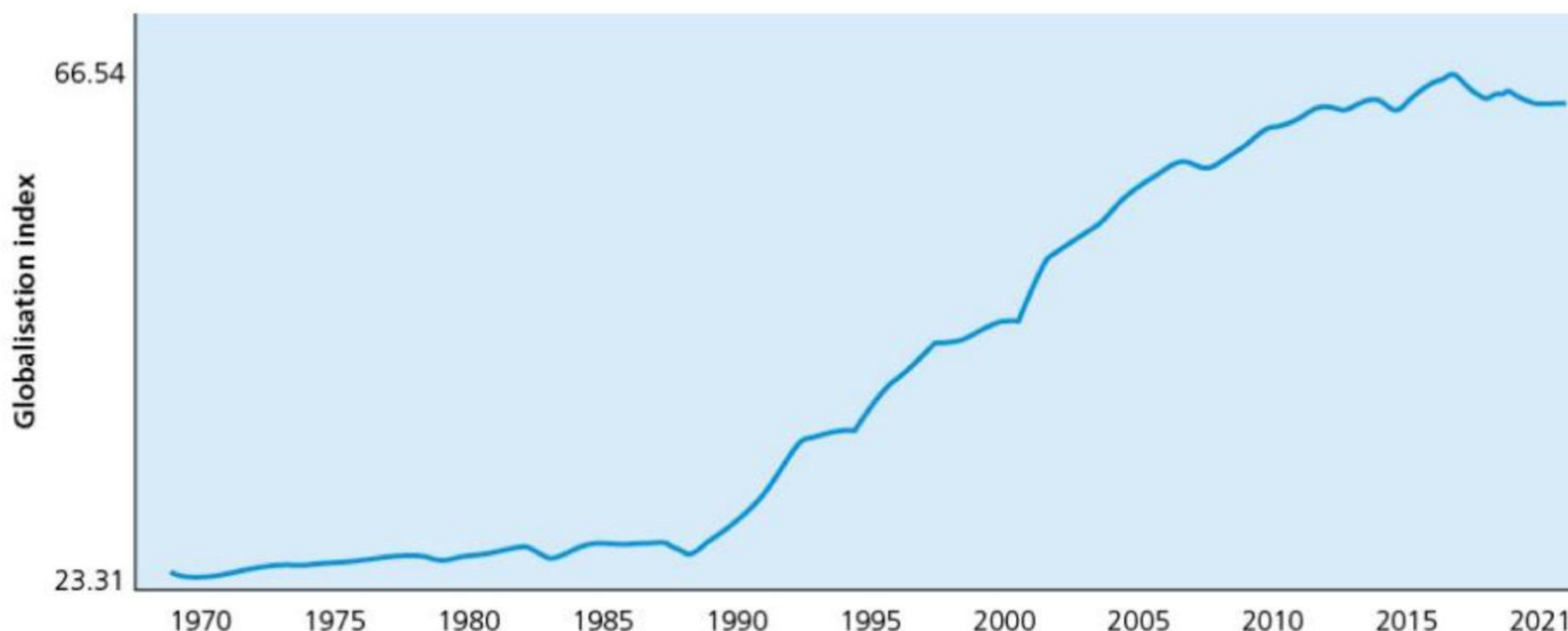
Like most countries, Mongolia has been impacted by globalisation. As a satellite state of the former Soviet Union, Mongolia was a communist country until after the collapse of the Soviet Union in 1989. Mongolia peacefully transitioned to an independent democracy in 1990, and in 1992 adopted a new constitution that established a free-market economy. Its membership of a wider range of international institutions followed, including:

- the World Trade Organization in 1997
- UN peacekeeping missions from 2002
- the International Organization for Migration in 2008.

Mongolia also established stronger relations with a number of major financial institutions, including the World Bank and the Asian Development Bank (ADB). The latter is Mongolia's largest multilateral development partner. In terms of bilateral aid (aid from a single donor country), Japan has been its major supporter ([Figure 9.30](#)). In 2019, ADB committed to a \$38 million loan to develop ecotourism in two national parks.



▲ **Figure 9.30** A Japanese bilateral aid project in Mongolia



▲ **Figure 9.31** The changes in Mongolia's KOF Globalisation Index, 1970–2021



▲ **Figure 9.32** A copper mine near Erdenet, northern Mongolia – copper is a major mineral resource in high demand around the world

The KOF Index of Globalisation

The KOF Index of Globalisation measures globalisation on economic, social and political dimensions. The index is on a scale of 1 to 100, with higher values denoting greater globalisation. [Figure 9.31](#) shows Mongolia's KOF Index from 1970 to 2021. The impact of globalisation on Mongolia was low while it was under the strong influence of the former Soviet

Union, as this to a large extent controlled the economy. The initial years of being an independent democracy were difficult as Mongolia had relied on economic assistance from Moscow for a long time. However, as Mongolia gradually opened up to the wider world economically, socially and politically, the country's increasing global connections encouraged incoming foreign direct investment. [Figure 9.31](#) shows the very significant change in Mongolia's globalisation index from the 1990s onwards.

Mongolia transitioned from an LIC to a lower-middle-income country in 2020. Its increase in GNI and other indicators of human development were mainly due to:

- FDI, which was almost entirely directed into the mining sector
- government investment funded by mineral royalties and external borrowing.

Economic characteristics

Mongolia relies heavily on the export of minerals. The valuable foreign exchange that this brings in allows the country to pay for the range of imports it requires. The country's extractive industries are based on large deposits of copper ([Figure 9.32](#)), gold, coal, tin, molybdenum, fluorspar, uranium and tungsten. Mongolia also has substantial known reserves of rare earth elements, but these have yet to be developed on any significant scale. The global production of rare earth elements is heavily dominated by China. Mongolia's known reserves add significantly to the country's strategic importance. The expansion of mining has led the transformation of the economy from its traditional dependence on herding and agriculture. The mining sector accounts for about 21 per cent of GDP.

The impact of Rio Tinto and the Oyu Tolgoi mine

Rio Tinto is a British-Australian TNC. It is recognised as the world's second-largest metals and mining company, after BHP. It operates in 35 countries and its revenue in 2023 amounted to US\$54 billion. In a strategic partnership with the government of Mongolia, Rio Tinto is currently developing the Oyu Tolgoi copper and gold mine in the southern Gobi region of the country ([Figure 9.33](#)). The mine is about 80 km north of Mongolia's border with China. It is the largest mining complex in Mongolia. Rio Tinto has 66 per cent ownership, and the government of Mongolia has the remaining 34 per cent. This form of joint ownership has become much more common in recent years in Mongolia in an attempt to ensure that the country obtains as fair a share as possible of the benefits of raw material exploitation in the country. Rio Tinto runs the mine on behalf of the partnership.

Oyu Tolgoi is one of the largest copper and gold deposits in the world. The mine is a combined open pit and underground mining project. The site was discovered in 2001, with construction beginning in 2010. Open pit mining began at Oyu Tolgoi in 2011. The copper concentrator, the largest industrial complex ever built in Mongolia, began processing mined ore into copper concentrate in 2013. The first batch of copper concentrate was shipped to customers in July 2013. In 2023, underground production began. This is where the greatest value of the project is located. At peak production, Oyu Tolgoi is expected to produce 500,000 tonnes of copper concentrate a year. When complete, the underground operation will have 200 km of tunnels, with the deepest shaft reaching 11.3 km. The project is using the leading underground mining technology known as 'block caving' to maximise the safety of employees. The project has been designed to be one of the most water-efficient copper mines in the world. The current infrastructure will allow the mine to operate for decades to come.



▲ **Figure 9.33** The Gobi Desert in south Mongolia

Socioeconomic benefits to Mongolia

As a partner in a joint venture, Mongolia is assured of a significant share of future profits, which are likely to be substantial. The global demand for copper is expected to remain very strong for many years to come as it is a material essential for decarbonisation and electrification. Oyu Tolgoi is expected to account for as much as 33 per cent of Mongolia's GDP once full commercial ore production starts.

In terms of value to the local and regional communities, a number of safeguards were negotiated between Rio Tinto and the national and regional government in Mongolia, including:

- Oyu Tolgoi's Cultural Heritage Management System, to ensure the management and protection of tangible and intangible cultural heritage
- a policy to create a supply chain in Mongolia with a particular focus on the southern Gobi region. This includes a 'Made

in Mongolia' procurement strategy to source products manufactured locally, such as miners' footwear. Oyu Tolgoi's spend in the southern Gobi region increased from \$0.5 million in 2010 to \$888 million in 2021.

Public opinion in Mongolia

Negotiations over the Oyu Tolgoi project were lengthy and subject to breakdown. A general feeling had developed in Mongolia over decades that:

- the country's natural resources were being exploited with insufficient benefits to the Mongolian population
- the environmental impact of mining operations was too high.

With an economy so reliant on mining, Mongolia had the difficult task of searching for the right balance between environmental conservation and natural resources development. It seems that while there was strong support for the economic opportunities associated with globalisation, there were significant concerns about the distribution of the benefits. Concerns about the fair distribution of economic gains were strongest in rural areas, where the 'trickle down' of wealth had been weaker and less obvious. There were concerns also about the erosion of traditional values and the spread of the worst aspects of Western culture.

Challenges ahead

- Mongolia continues to experience its three main geographical challenges – that it is landlocked, remote and vast (although to a lesser extent than in the past). These issues are compounded by infrastructure and logistics bottlenecks.
- Mongolia has attempted to maintain good relations with its two giant neighbours, Russia and China, as well as having a good dialogue with the USA and other countries of strategic importance. The China–Mongolia–Russia Economic Corridor has become an important element of China's Belt and Road Initiative and a major component of cooperation between China and Russia.
- With such extremes of climate and environment, Mongolia is very sensitive to climate change. Mongolian scientists say that the average temperature has increased by over 2°C since 1940.

Activities

- 1 Describe the geographical location of Mongolia.
 - 2 Study [Figure 9.31 \(page 230\)](#). What were the reasons for the increase in Mongolia's KOF Globalisation Index from the early 1990s?
 - 3 Write a bulletpoint summary of the ownership and development of the Oyu Tolgoi mine.
 - 4 How important is revenue from the mine expected to be when it is fully operational?
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